A Practical PLN-RAG Extension for Identity-Aware Semantic Parsing

A (Relatively) Simple Path Implementing the Essential PGM-HoTT and TransWeave

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Abstract

The current `canonical_pln` parser is already the strongest configuration in the PLN-RAG benchmark experiments recently run, but its remaining failures cluster around a single underlying weakness: it treats every mention of a concept as the same symbol, regardless of which prototype the speaker had in mind, which sentence or chunk the mention came from, and which other mentions might (or might not) refer to the same entity. This note proposes a focused extension, `canonical_senf_pln`, that sits between canonicalization and query planning. It tracks frames, roles, entity mentions, exemplar distances, graded identity evidence, and weighted cross-context alignments. These are the practical content of prior theoretical proposals involving PGM-HoTT and TransWeave, which we here indicate can be implemented as ordinary records, weighted graphs, top-k search, and PLN truth-value adjustment. The alignment layer is structured hierarchically: a cheap heuristic mapping serves as a pushforward seed, and a Sinkhorn-style local-to-global polish corrects it where the budget pays for itself. This view turns the parser's existing deterministic work (canonical kind matches, lexical aliases, anaphora cues) into the seed of the new layer rather than a competitor to it.

The presentation is kept intentionally elementary here. No HoTT proof terms, no univalence machinery, no infinity-groupoids. The HoTT material is treated as the mathematical explanation for why the engineering layer takes this shape; the appendix gives the correspondence for interested implementers.

AI Assistance Note A bunch of help from GPT-5.5-Pro and Claude Opus 4.7 here

1 The problem this layer solves

A benchmark summary recently published from the current PLN-oriented semantic parsing pipeline shows that `canonical_pln` produced 13 proofs out of 25 cases at 52% proof rate, well ahead of the nl2pln baseline (8%) and ahead of `canonical_langextract` (36%). The remaining failures are not random. They cluster around three recurring patterns.

Pattern A: same word, different referents. A chunk says "the camera" in one sentence and "their camera" two sentences later. Canonicalization gives both the symbol `camera`. The reasoner then either treats them as the same instance (sometimes wrong) or proves nothing (sometimes because the relevant fact was attached to the other mention).

Pattern B: same symbol, different prototypes. A rule learned about "games requiring strategy" (chess-like) is applied to a query about "a game that left me exhausted" (football-like) because the symbol `Game` matches in both. The truth value does not degrade. The proof goes through with unjustified confidence.

Pattern C: contradictory evidence collapses the chain. A passage contains a claim from one source that conflicts with a claim from another. Symbolic generation produces both atoms. The reasoner either silently picks one or fails to bind. There is no way to keep both and propagate uncertainty.

These three patterns share a common cause: canonical PLN symbols are correct as far as they go, but they discard enough information about *which instance, which prototype, and which source* that the reasoner cannot recover the relevant distinctions. A pure NL2PLN reasoner cannot retrieve those distinctions either, because it does not represent them.

The proposed layer recovers exactly those three things and exposes them to the query planner and to PLN.

2 Implementation slogan

While my previous suggestions for working around these sorts of issues were mathematical in expression, in practice the extension we need to solve the practical problems should not begin with proof terms, univalence, or a full HoTT interpreter. It should begin with the following engineering object:

> A costed identity-and-exemplar layer on top of `canonical_pln`, producing weighted TransWeave alignments and PLN bridge links.

The layer answers four operational questions that ordinary symbolic parsing tends to answer too rigidly:

1. Which mentions probably refer to the same thing?
2. Which evidence says they probably do not refer to the same thing?
3. Which exemplar or prototype of a kind is active in this context?
4. How much confidence should be lost when a fact or rule is transported across a paraphrase, a granularity shift, an analogy, or a new chunk?

The HoTT documents give a mathematical foundation for these questions. The implementation can use a simpler operational form:

HoTT-side idea	Practical object	First implementation
Graded identity paths	Identity edge	`IdPlus` and `IdMinus` scores
Multi-exemplar type	Exemplar registry	Small exemplar sets per kind
Definite description	Exemplar-aware selection	Resolver for `the X`, pronouns, aliases
Transport	TransWeave	Weighted alignment between SENFs
Paraconsistency	Guarded alternatives	Keep ranked conflicting hypotheses
Modal/HoTT operationalization	MeTTa-IL later	Optional cost and bilattice syntax

The right-hand column is the only part the implementation team needs to build.

3 Where this attaches to the existing PLN-RAG parser

The existing parser already has most of the required outer architecture. The new extension fits in a single well-defined window: after canonical PLN generation and canonicalization, but before query planning and final reasoner execution. Inserting it earlier would interfere with symbol normalization. Inserting it later would deprive the query planner of the new signals.

Natural language input
↓
existing concept extraction and parser hints
↓
existing NL2PLN generation
↓
existing canonicalization and structural filtering
↓
new SENF builder
↓
new exemplar scorer and identity graph builder
↓
new TransWeave aligner across sentences and chunks
↓
new PLN bridge generator
↓
existing query planning, now with identity and exemplar costs
↓
executable symbolic output

This is deliberately conservative. It preserves the central lesson of the PLN-RAG workstream: proof quality depends heavily on canonical symbolic reuse, structural alignment, query executability, and parser/reasoner contract enforcement. The new layer adds better semantic alignment signals; it does not replace the working canonicalization and query planning machinery. In particular, the parser/reasoner contract continues to hold: invalid symbolic structures are rejected, not silently mutated. The new layer's outputs go through the same contract.

4 Semantic Elegant Normal Form as the practical intermediate layer

The HoTT-side material refers to SENF as a typed normal form for natural-language meaning. The implementation should treat SENF as an intermediate representation between canonical PLN atoms and query planning.

Each SENF field exists for a specific operational reason. Before giving the signature, it is worth being explicit about what each field buys:

- *Frames* are needed because the events and situations a sentence describes are what carry roles. A predicate alone ("lend") cannot bind an agent and a recipient; a frame can.
- *Entities* are needed because the reasoner needs to talk about specific mentions, not just kinds. "Alex" and "Sam" are two distinct entities, both of kind `Person`.

- *Kinds* are the canonical PLN side: `Camera`, `Game`, `Lens`. These are already produced by `canonical_pln`.
- *Exemplars* are new. They record which prototype of a kind is active in this context âĂŤ chess versus football for `Game`, professional versus phone for `Camera`.
- *Roles* attach entities to frames with labels such as agent, theme, goal.
- *Constraints* carry the things that do not fit cleanly as roles: definiteness, uniqueness, time, location, negation, modality.
- *Identity graph* records which mentions probably refer to the same thing and which probably do not. It is the heart of the new layer.

A SENF object then contains:

$$S = (F, E, K, X, R, C, G),$$

with the field meanings above. A minimal signature for the assertions inside is:

$$\begin{aligned}
\mathsf{Frame}(f,h) &\quad \text{frame } f \text{ has head } h,\\
\mathsf{Role}(f,r,e) &\quad \text{entity } e \text{ fills role } r \text{ in frame } f,\\
\mathsf{isa}(e,k) &\quad e \text{ is of kind } k,\\
\mathsf{exemplar}(e,k,x,d) &\quad e \text{ is distance } d \text{ from exemplar } x \text{ of kind } k,\\
\mathsf{nearest_ex}(e,k,x) &\quad x \text{ is the nearest active exemplar for } e \text{ as kind } k,\\
\mathsf{sel}(p,e) &\quad e \text{ is selected as the contextual witness of predicate } p,\\
\mathsf{uniq}(p) &\quad p \text{ is treated as contextually unique},\\
\mathsf{Id}^{+}(e,e',c,r) &\quad \text{positive evidence for sameness, cost } c,\ \text{reasons } r,\\
\mathsf{Id}^{-}(e,e',c,r) &\quad \text{positive evidence for difference, cost } c,\ \text{reasons } r.
\end{aligned}$$

The single most important design decision is that the system stores *both* positive sameness evidence and positive difference evidence on the same edge. A contradiction does not erase an identity hypothesis. It adds a negative channel, lowers confidence, and possibly creates a guarded alternative. This is what makes the layer paraconsistent in practice.

4.1 Example SENF fragment

For the sentence

> Alex lent Sam the Nikon camera.

one possible quasi-MeTTa SENF representation is:

```
(Frame s1_f1 lend)
(Role s1_f1 agent alex)
(Role s1_f1 goal sam)
(Role s1_f1 theme cam1)
(isa cam1 Camera)
(has-prop cam1 Nikon)
```

```
(sel (and (isa $x Camera) (has-prop $x Nikon)) cam1)
(uniq (and (isa $x Camera) (has-prop $x Nikon)))
(exemplar cam1 Camera professional_camera 0.12)
(exemplar cam1 Camera consumer_camera 0.48)
(nearest-ex cam1 Camera professional_camera)
```

A few details are worth noticing:

- The frame `s1_f1` groups the three roles, so the reasoner can talk about the lending event as a whole.
- The entity `cam1` is distinct from the kind `Camera`. The next sentence's "it" will get a fresh entity (`it_s2`); the identity graph will then propose `cam1 = it_s2` with a cost and reasons.
- The two `exemplar` lines record graded membership in two prototypes. The 0.12 vs. 0.48 spread comes from the "Nikon" cue: a Nikon is much more typical of professional cameras than of consumer cameras.
- The `nearest-ex` fact is the convenient summary the rest of the system actually queries.

The implementation does not need to solve a large ontology problem at first. It only needs enough exemplar structure to improve identity and query alignment on the cases that currently fail.

5 The exemplar registry

Start with a small exemplar registry for common and domain-relevant kinds:

```
(ex-set Camera (professional_camera consumer_camera phone_camera security_camera))
(ex-set Game (chess_game football_game childrens_game video_game))
(ex-set Bird (robin eagle penguin ostrich))
(ex-set Treatment (drug_treatment surgical_treatment behavioral_treatment))
```

Each entity mention gets a small vector of distances to the exemplars of its kind. These distances come from any of: lexical cues ("Nikon" shortens distance to `professional_camera`), local context ("strategy" shortens distance to `chess_game`), retrieval metadata, embedding similarities, manually assigned domain hints, or LLM judgments constrained by canonical symbols.

For an entity e of kind k, let X_k be the exemplar set of k. Store:

$$d_X(e, x) \in [0, 1] \quad \text{for each } x \in X_k.$$

Then the nearest active exemplar is:

$$\mathsf{nearest\_ex}(e, k) = \operatorname*{argmin}_{x \in X_k} d_X(e, x).$$

When several exemplars are nearly tied, keep them under guards rather than forcing a premature choice:

$$\mathsf{AltEx}(e, k) = \{x \in X_k : d_X(e, x) \leq d_{\min}(e, k) + \tau_X\}.$$

A concrete reading: suppose "the game" appears in a chunk and produces $d_X(\text{gameA}, \text{chess\_game}) = 0.05$, $d_X(\text{gameA}, \text{football\_game}) = 0.80$, $d_X(\text{gameA}, \text{childrens\_game}) = 0.60$. With $\tau_X = 0.10$, the alternative set AltEx is the singleton $\{\text{chess\_game}\}$: the chess prototype is clearly active and no alternative is held open. If instead the distances were $0.05, 0.10, 0.55$, then both chess and football would stay under a guard, and the eventual choice would depend on downstream evidence.

This directly implements the idea that a word such as "game" invokes the chess exemplar in a strategy context and the football exemplar in an exhausting-activity context.

6 Identity graph

For each chunk or sentence window, build an identity graph whose vertices are entity mentions and whose edges are identity hypotheses.

A practical identity edge has the form:

$$I(e, e') = (e, e', c^+, c^-, R^+, R^-, g),$$

where:

- c^+ is the cost of identifying e and e'. Lower is better.
- c^- is the cost or score associated with evidence for non-identity. The implementation may store this either as a cost-to-refute (cheap evidence for difference $\Rightarrow$ low c^- $\Rightarrow$ strong penalty) or as a contradiction strength (high value $\Rightarrow$ strong penalty). The convention must be explicit; this note uses cost-to-refute.
- R^+ is a set of reasons supporting sameness.
- R^- is a set of reasons supporting difference.
- g is a guard: sentence id, chunk id, source id, time interval, modality, or speaker.

A simple scoring convention is:

$$C_{id}(e, e') = w_+ c^+(e, e') + w_- p^-(e, e') + w_g C_{guard}(e, e'),$$

where p^- is a penalty derived from the negative identity channel. With the cost-to-refute convention,

$$p^-(e, e') = \frac{1}{1 + c^-(e, e')}.$$

Cheap evidence for difference (small c^-) gives a high penalty close to 1; expensive evidence for difference (large c^-) gives a small penalty close to 0.

Worked numeric example. Take the camera case. Suppose for the pair (`cam1`, `it_s2`) the system finds $c^+ = 0.15$ (cheap to identify: anaphora, theme-role match, return-after-lend pattern), $c^- = 0.80$ (it would cost a lot to refute, only a weak time-shift cue), $w_+ = w_- = 1.0$, $w_g = 0$. Then

$$p^- = \frac{1}{1+0.80} \approx 0.56, \qquad C_{id} = 1.0 \cdot 0.15 + 1.0 \cdot 0.56 + 0 \approx 0.71.$$

A different pair might invert this. For (`lens1`, `lens_claim_s4`) in the Casey example, $c^+ = 0.42$ (same kind, same topic, same person Sam) and $c^- = 0.21$ (the exemplar shift from camera-lens to standalone-lens is cheap evidence for difference, plus an explicit conflicting claim). Then

$$p^- = \frac{1}{1+0.21} \approx 0.83, \qquad C_{id} = 0.42 + 0.83 \approx 1.25.$$

The second pair is roughly twice as costly to treat as the same entity. The query planner should treat the first as a near-certain identity and the second as a guarded alternative.

6.1 Reasons that should contribute to identity scores

Positive identity reasons include:

- lexical match or canonical alias match;
- same canonical kind;
- compatible role position;
- anaphoric cue, such as a nearby pronoun;
- definite-description cue, such as `the camera`;
- compatible time and location granularity;
- close exemplar match;
- same source entity id or retrieval grounding.

Negative identity reasons include:

- incompatible kind;
- incompatible time or simultaneous-location conflict;
- incompatible role pattern;
- incompatible exemplar match;
- explicit contrastive language;
- conflicting speaker or source claim;
- incompatible part-whole relation.

The system should keep the reasons because they provide useful explanations and debugging information, and they let the eventual evaluation harness distinguish well-grounded identifications from lucky guesses.

7 What a TransWeave is: concept and mathematics

The identity graph in the previous section connects mentions inside a single chunk. A TransWeave generalizes this idea to a complete alignment between two SENFs — two sentences, two chunks, a retrieved passage and a query representation, or a source-rule context and a target context where the rule is being applied.

This section gives the conceptual picture, two complementary readings of the mathematical object, the typed components, and the cost-trinity that organizes the implementation. The detailed cost formulas and algorithms come in the next section.

7.1 The concept

A TransWeave between SENFs S_A and S_B is a structured, costed correspondence that does three things:

1. Says which pieces of S_A correspond to which pieces of S_B — entities to entities, frames to frames, kinds to kinds, exemplars to exemplars, roles to roles, time grains to time grains, location grains to location grains.

2. Records how much structural distortion the correspondence introduces. A weave that preserves all pairwise relationships is a literal identity. A weave that scrambles them is an analogy or a paraphrase with shifted role structure.

3. Accumulates a total transport cost that the PLN bridge generator uses to degrade transported truth values.

A TransWeave is not a synonym list and not a string-match map. It is a typed alignment of semantic structure, with the typing of frames, roles, entities, and exemplars all respected by the alignment.

7.2 Two readings of the same object

Two ways of looking at a TransWeave are useful in practice. Both describe the same underlying object; they differ in which aspects they emphasize.

The morphism reading. A TransWeave is a partial typed morphism between SENF structures. In the HoTT picture from the PGM-HoTT note, a SENF is an inhabitant of a semantic fiber $\mathsf{Sem}(c)$ over a context c, and a weave $W : \mathsf{Sem}(c_A) \to \mathsf{Sem}(c_B)$ is approximately a graded equivalence between fibers that can be transported across via a guarded form of univalence. The implementation does not need any of this machinery, but the morphism reading is what justifies treating a weave as a typed object: its ϕ_* components must respect the typing of the corresponding SENF fields, not just match strings.

The coupling reading. A TransWeave is a (possibly soft) coupling between the elements of S_A and S_B. Each partial map ϕ_* corresponds to a coupling matrix Π_* over the source and target elements of the appropriate field:

$$\Pi_* \geq 0, \qquad \Pi_* \mathbf{1} \leq \mu_A, \qquad \Pi_*^\top \mathbf{1} \leq \mu_B,$$

where μ_A and μ_B are mass distributions over source and target elements — uniform in the simplest case, weighted by salience or parser confidence in richer versions. This reading is the bridge to optimal-transport machinery, including the hierarchical Sinkhorn approach in Âğ8.4. It is also the reading the implementation should use when the SENFs are large and a single hard assignment is too brittle.

A *hard weave* is the special case where each Π_* has at most one nonzero entry per row and per column — each source element maps to one target element or to nothing. A *soft weave* spreads mass across several candidates, keeping ambiguity until later stages resolve it. Both forms have the same typed-tuple structure; they differ only in the sparsity of the underlying couplings.

7.3 The defining condition: Bellman commutation

The two readings above describe what a TransWeave *looks like* (a typed tuple, a coupling). They do not say what makes the object a *weave* rather than an arbitrary structured map. The missing piece is the commutation condition.

Each context c comes with a dynamical operator $B_c : \mathsf{Sem}(c) \to \mathsf{Sem}(c)$ that propagates SENF state by one step: one sentence to the next, one PLN inference step to the next, one retrieval chunk to the next. The specific form of B depends on what counts as a step (appendix B gives the precise version); the conceptual content is that each context has its own way of evolving meaning forward.

A mapping $W : \mathsf{Sem}(c_A) \to \mathsf{Sem}(c_B)$ is a TransWeave with intertwining error ε when

$$\|W \circ B_A - B_B \circ W\| \le \varepsilon$$

in an appropriate norm. In plain English: doing a step of reasoning in A and then weaving over to B produces (up to ε) the same SENF state as weaving over first and then doing a step of reasoning in B. This is the Bellman-Darboux intertwining condition adapted to discourse, and it is what justifies calling W a *weave* — it weaves together two dynamical systems by approximately commuting with their evolution operators. A correspondence that did not commute would not be a weave; it would just be a renaming.

The intertwining error ε is what the weave cost $C(W)$ ultimately bounds. The exponential confidence-degradation formula in Âğ8.5 is essentially passing ε through to truth-value discounting: large ε means large $C(W)$ means heavily discounted s'. The hierarchical Sinkhorn polish in Âğ8.4 is what drives ε down, one scale at a time. Appendix B treats the commutation condition in full and shows why the marginal constraints in the polish stages are equivalent to enforcing it approximately.

7.4 The components: a typed tuple

A TransWeave between S_A and S_B is a tuple

$$W : S_A \rightsquigarrow S_B, \qquad W = (\phi_F, \phi_E, \phi_K, \phi_X, \phi_R, \lambda_t, \lambda_l, \rho, C),$$

where:

- ϕ_F maps frames in S_A to frames in S_B (hard or soft).
- ϕ_E maps entities in S_A to entities in S_B.
- ϕ_K maps kinds in S_A to kinds in S_B.

- ϕ_X maps active exemplars in S_A to active exemplars in S_B.
- ϕ_R maps role labels or role patterns.
- λ_t aligns time granularities.
- λ_l aligns location granularities.
- ρ records distortion: the maximum change in pairwise structural distance.
- C is the total transport cost.

Downstream consumers of the weave read different fields: the PLN bridge generator reads C and the per-component costs to decide how much truth value to discount; the query planner reads ϕ_E and ϕ_X to find executable bindings; the diagnostics layer reads R^+, R^- on the underlying identity edges to explain why a particular alignment was selected; the test harness reads ρ to flag weaves that are cheap on average but locally distorted.

7.5 The three transport-cost regimes

The total cost $C(W)$ aggregates contributions from many small terms (see Âğ8.1 for the full list). It is useful to group those terms into three families that mirror the transport-trinity from the PGM-HoTT note:

- *Structural cost* $C_{struct}(W)$ — how much does the weave change the relational structure? This is a Gromov-Wasserstein-like term: do mapped entities preserve their pairwise role-and-time distances?
- *Distributional / exemplar cost* $C_{dist}(W)$ — how far apart are the active exemplars under the map? This is a Wasserstein-like term over the exemplar metric d_X.
- *Conflict / dynamic cost* $C_{conf}(W)$ — how much negative identity evidence does the weave have to absorb? This term is large when the IdMinus channels carry strong evidence for difference.

The decomposition is useful because the relative sizes of these terms tell the diagnostics layer what kind of mismatch dominated:

- Low structural, high exemplar: **analogy** (same shape, different prototype) — the chess-to-football rule transfer case.
- Low exemplar, high structural: **paraphrase with scrambled roles** (same prototypes, different argument structure).
- Low structural and exemplar, high conflict: **disputed referent** (good alignment but conflicting sources) — the camera-lens vs. standalone-lens case in Âğ9.
- All three low: **literal identity** (transport with little confidence loss).
- All three high: **weak alignment** (the planner should treat this as a guarded alternative or reject it).

This taxonomy is what makes the cost structure useful for the rest of the system. The transport cost is not a single number that drives confidence degradation; it is a typed signature of the kind of transport that occurred.

8 TransWeave in practice

This section gives the cost formulas, algorithms, and PLN-bridge generation that turn the conceptual picture into working code. The components $\phi_F, \phi_E, \phi_K, \phi_X, \phi_R, \lambda_t, \lambda_l, \rho, C$ are as defined in Âğ7.3.

8.1 Cost components

For an entity pair (e, e'), use a cost such as:

$$\begin{aligned} C_E(e, e') = a_1 C_{name}(e, e') + a_2 C_{kind}(e, e') + a_3 C_{role}(e, e') + a_4 C_{time}(e, e') \\ + a_5 C_{loc}(e, e') + a_6 C_{ex}(e, e') + a_7 C_{id}(e, e') + a_8 C_{source}(e, e'). \end{aligned}$$

For a frame pair (f, f'):

$$\begin{aligned} C_F(f, f') = b_1 C_{head}(f, f') + b_2 C_{roles}(f, f') + b_3 C_{args}(f, f') + b_4 C_{polarity}(f, f') \\ + b_5 C_{time}(f, f') + b_6 C_{loc}(f, f') + b_7 C_{modal}(f, f'). \end{aligned}$$

For exemplar alignment:

$$C_X(x, x') = d_X(x, x'),$$

or with a generic-exemplar route:

$$C_X^*(x, x') = \min\big(d_X(x, x'), d_X(x, x_{gen}) + d_X(x_{gen}, x') + \delta_{gen}\big).$$

The generic-exemplar route is useful when direct exemplar alignment is too strong but both entities still belong to the same broader kind. For example, chess and football are far apart as exemplars of `Game`, but both align cheaply through a generic-game exemplar. This is what lets the system express “these are both games, but treat the analogy as weak” rather than forcing a binary choice between “same exemplar” and “no alignment”.

The full weave cost is:

$$\begin{aligned} C(W) = \sum_{(f,f') \in \phi_F} C_F(f, f') + \sum_{(e,e') \in \phi_E} C_E(e, e') + \sum_{(x,x') \in \phi_X} C_X^*(x, x') \\ + \lambda_u U(W) + \lambda_d \rho(W) + \lambda_c C_{conflict}(W), \end{aligned}$$

where $U(W)$ penalizes unmatched but important structure, $\rho(W)$ penalizes distortion, and $C_{conflict}(W)$ penalizes negative identity evidence.

8.2 Distortion

The distortion term checks whether the weave preserves the internal geometry of the source SENF. Let d_A and d_B be structural or exemplar distances inside the two SENFs. For mapped items u, v in S_A,

$$\rho(W) = \max_{u,v} |d_A(u,v) - d_B(\phi(u), \phi(v))| \,.$$

A weave with low total cost but high distortion is a warning sign: the alignment is locally cheap for each pair, but it scrambles the global structure. Such weaves should be rejected as literal identity and treated only as analogies with degraded confidence.

8.3 Single-shot TransWeave algorithm: a baseline

The algorithm below is deliberately generic. The implementation may use Hungarian matching, beam search, local search, or an ILP solver, but the contract is the same: produce the top k weighted weaves rather than only the single best alignment. This is the right approach for small SENFs (one or two sentences each); the hierarchical version in Âğ8.4 is the recommended approach for longer texts and rule transfer.

Algorithm 1 Build top-k TransWeaves between two SENFs (single-shot)

Require: Source SENF S_A, target SENF S_B, beam width k, thresholds τ_{id} and τ_{ex}
Ensure: Ranked guarded weaves $W_1, \ldots, W_k$

```
Extract frames, entities, roles, kinds, exemplars, times, and locations from S_A and S_B.
Construct candidate frame pairs using head, role, time, location, and polarity compatibility.
Construct candidate entity pairs using kind, role position, lexical alias, anaphora, and source grounding.
Score exemplar pairs and generic-exemplar routes for each candidate entity pair.
Add positive identity support and negative identity penalties to each candidate entity pair.
Search over compatible frame and entity assignments.
for all candidate assignment A in the search frontier do
    Complete role maps, time maps, location maps, and exemplar maps.
    Compute total cost C(A) and distortion ρ(A).
    Mark close alternatives as guarded rather than deleting them.
end for
Return the top k assignments as weaves, ranked by total cost and distortion.
```

8.4 Hierarchical TransWeave: heuristic seed plus iterative polish

For longer texts, multi-chunk RAG, or rule transfer where the source and target SENFs each contain many frames and entities, single-shot optimization is both slow and poorly conditioned. A cleaner approach mirrors the structure of dynamic programming and the hierarchical TransWeave used in coordination-problem transfer [5]: build a cheap heuristic seed first, polish it locally where it is wrong, propagate corrections upward into larger blocks, and finish with a small global polish.

This is also the cleanest way to explain the relationship between the parser's existing deterministic work and the new TransWeave layer. The parser already produces a usable heuristic mapping — canonical kind matches, lexical aliases, anaphora cues, role-position correspondences. The hierarchical view says: TransWeave does not throw that work away. It uses it as the pushforward seed, and the polish stages correct it where it is wrong.

The four levels.

1. **Heuristic pushforward (seed).** A fast deterministic mapping built from signals the parser already has: canonical kind matches, lexical aliases, anaphora cues, role-position correspondences, source-grounding from retrieval, exemplar nearest-neighbors. This is the pushforward $T_X\#S_A$ — cheap, often distorted at the margins, but in roughly the right place.

2. **Local polish (per pair, per small frame).** For each candidate pair (entity-pair or single-frame-pair), run a few iterations of marginal-matching that enforce local consistency: role-marginal feasibility (if A's `theme` maps to B's `theme`, their fillers should agree on kind), exemplar coherence, identity-graph compatibility. The Sinkhorn / IPF structure from entropic optimal transport gives the cleanest update rule, but any local fixed-point iteration that enforces row and column sums works. Keep the number of sweeps small: $k_0 = 2$–5.

3. **Block polish (small subgraph).** Group nearby frames into blocks of width W — for example, a main frame plus its argument frames, or a small sentence window. Run a block-level update that enforces consistency at block boundaries: an entity that appears in two adjacent frames must map consistently in both. This is where corrections from one local pair are reconciled with corrections from an adjacent local pair. Grow W by doubling until it covers the whole SENF pair.

4. **Global polish (full SENF pair).** A small number of sweeps ($k_2 = 1$–3) over the whole weave to enforce constraints that the local and block levels could not see: end-of-chunk anaphora resolution, top-level rule-context constraints, query-binding feasibility.

Each level enforces commutation at a different scale. The hierarchical structure is not an arbitrary computational convenience. As Âğ7.4 noted, a TransWeave is defined by the Bellman commutation $\|W \circ B_A - B_B \circ W\| \leq \varepsilon$. The marginal constraints inside Sinkhorn/IPF sweeps are exactly what enforce this commutation approximately: local polish drives down the residual at single-step scale (one B application per pair), block polish drives it down at the block-width scale (W applications of B), global polish drives it down at the horizon scale (full discourse). Appendix B works through this correspondence; for the implementation, the practical point is that the polish residuals at each level are direct estimates of ε at that scale, and they are exactly the right diagnostic signal for "this weave is well-formed" vs. "this weave is being forced."

Why hierarchy helps in the parser setting.

- *Locality of natural-language structure.* Most identity and role decisions are local: a pronoun usually resolves within a sentence or two, an exemplar choice is set by the nearest cue, role fillings are governed by their immediate frame. The hierarchical structure puts the polish work where the evidence is.

- *Bellman / DP-style decomposition.* Weave costs decompose additively over frames and pairs. Local optima first, block consistency next, global only at the end — exactly the structure of dynamic programming.

- *Numerical conditioning.* A single global optimization over a large SENF pair has many tight constraints simultaneously. Doing local sweeps first makes the global stage much better conditioned, so it converges in a handful of iterations rather than dozens.

- *Diagnostic transparency.* When a weave is bad, the hierarchical structure makes it easy to localize where it went wrong — which local pair, which block boundary, which global constraint. A single-shot optimization gives a single bad number with no decomposition.

Algorithm 2 Hierarchical TransWeave for SENF pairs

Require: Source SENF S_A, target SENF S_B, transfer hints T, regularization ε, sweep counts k_0, k_1, k_2
Ensure: Refined weave W and diagnostics

// **Step A: heuristic pushforward seed**
Propose $\Pi_E^{(0)}$ from canonical kind matches, lexical aliases, anaphora, role positions, retrieval grounding.
Propose $\Pi_F^{(0)}$ from head match, role pattern, time and location compatibility.
Push exemplars through under $\phi_X^{(0)}$ using d_X.
// **Step B: local polish**
for each candidate entity pair (e, e') in parallel **do**
    Run k_0 Sinkhorn/IPF sweeps enforcing role-marginal consistency.
end for
for each candidate frame pair (f, f') in parallel **do**
    Run k_0 sweeps against argument marginals.
end for
// **Step C: block polish**
$W \leftarrow 1$ ▷ initial block: one frame plus its arguments
while $W < |F_A|$ **do**
    for each block of W frames in parallel **do**
        Polish block boundaries for identity and exemplar consistency (k_1 sweeps).
        Redistribute corrections back to the constituent local couplings.
    end for
    $W \leftarrow \min(2W, |F_A|)$
end while
// **Step D: global polish**
Run k_2 sweeps over the full SENF pair, enforcing terminal and query constraints.
// **Step E: emit weave**
Assemble $W = (\phi_F, \phi_E, \phi_K, \phi_X, \phi_R, \lambda_t, \lambda_l, \rho, C)$ from the polished couplings.
return the top-k weaves ranked by C and ρ.

Algorithm.

Practical knobs. A few options worth exposing:

- *Skip levels.* For small SENFs (one or two frames each), skip the block stage. For very large ones, allow W to grow further before global polish.
- *Multiple seeds.* Initialize from several different heuristic seeds — anaphora-first, role-first, kind-first — and keep the top-k polished weaves. The polish stage is cheap enough that multi-seed runs are affordable.
- *Adaptive regularization.* Use higher entropy regularization ε in early local sweeps (soft mappings that explore alternatives), lower it in block and global sweeps (sharper mappings).

- *Truncated propagation.* Cap the depth of block aggregation when budget is tight. The system degrades to a local-only weave, still useful for short texts.
- *Hot starts.* For repeated rule transfer between similar contexts (the same chess-vs-football pattern across many sentences), cache the heuristic seeds and the polished local couplings.

Walking through the worked examples. The hierarchical reading clarifies what happens in the examples of Âğ9. The pronoun-camera weave (final cost 0.20) breaks down like this. The heuristic seed maps `it_s2` to `cam1` from canonical kind match, theme-role match, and recency cue: the cheap pushforward. Local polish verifies that the exemplar mapping (`professional_camera` to `professional_camera`) is feasible under role-marginal consistency, and tightens the cost slightly. No block polish is needed for a two-sentence span. A single global sweep checks that the terminal constraint (Sam being the agent of the return frame) is satisfied. Final cost: 0.20.

The chess-to-football weave (cost 0.91) reveals its high cost *at the heuristic seed stage already*: the seed produces a high exemplar shift ($d_X(\text{chess}, \text{football}) = 0.80$). Local polish cannot reduce it, because the exemplars are genuinely different. Block and global polish leave the cost essentially where it was. This pattern — hierarchy reveals quickly which weaves are cheap and which are forced — is the main diagnostic advantage of the hierarchical structure. A single-shot optimization would have produced the same final cost but with no early signal that the dominant contribution was the exemplar shift.

8.5 Confidence transport into PLN

A weave should produce bridge atoms for PLN. A bridge atom can express similarity, inheritance, role transfer, or rule transfer. Its truth value should be degraded by the transport cost.

If a source fact has PLN truth value (s, w) and is transported through a weave W with total cost $C(W)$:

$$\begin{aligned} s' &= s \exp(-\lambda_s C(W)), \\ w' &= \frac{w}{1 + \lambda_w C(W)}. \end{aligned}$$

Worked numeric example. Take a source rule with $(s, w) = (0.85, 0.70)$ and a high-cost weave with $C(W) = 0.91$ (such as a direct chess-to-football mapping, computed below). With $\lambda_s = 1.0$ and $\lambda_w = 1.0$,

$$s' = 0.85 \cdot e^{-0.91} \approx 0.85 \cdot 0.403 \approx 0.34, \qquad w' = \frac{0.70}{1.91} \approx 0.37.$$

A lower-cost weave with $C(W) = 0.20$ (such as the pronoun case below) gives

$$s' = 0.85 \cdot e^{-0.20} \approx 0.85 \cdot 0.819 \approx 0.70, \qquad w' = \frac{0.70}{1.20} \approx 0.58.$$

The same source rule transports almost intact through a clean pronoun alignment but loses most of its strength when forced across an exemplar shift.

If the implementation wants to separate exemplar cost from other costs, use:

$$s' = s \exp(-\lambda_X C_X(W)) \exp(-\lambda_R C_{role}(W)) \exp(-\lambda_C C_{conflict}(W)).$$

This is intentionally simple. The first implementation should privilege clear calibration over mathematical sophistication; the exponential form is convenient because each cost component contributes multiplicatively and the values stay in $[0, 1]$.

8.6 Quasi-MeTTa representation of a weave

```
(Weave W12 S1 S2)
(WeaveCost W12 0.23)
(WeaveDistortion W12 0.07)

(MapFrame W12 s1_f1 s2_f1)
(MapEntity W12 cam1 it_s2)
(MapRole W12 theme theme)
(MapKind W12 Camera Camera)
(MapExemplar W12 professional_camera professional_camera)

(IdPlus cam1 it_s2 0.18
  (reasons anaphora-after-lend theme-role-match exemplar-match))

(IdMinus cam1 it_s2 0.74
  (reasons time-shift underspecified-pronoun))

(SimilarityLink cam1 it_s2 (stv 0.84 0.72))
(ContextLink (Exemplar Camera professional_camera)
  (SimilarityLink cam1 it_s2 (stv 0.84 0.72)))
```

The exact surface syntax is not important at first. What matters is the presence of a typed alignment object that the query planner can inspect, the test harness can score, and the diagnostics layer can explain.

9 Worked TransWeave examples

9.1 Example 1: pronoun and definite-description alignment

Input:

S1: Alex lent Sam the Nikon camera.
S2: Sam returned it.
S3: Alex found their camera at home. The lens was cracked.
S4: Casey claims Sam borrowed the lens on Monday.

The desired behavior is not to force one global identity decision. The system should produce several guarded identity hypotheses with appropriately ranked costs.

The four SENFs. Canonicalization yields the following structures, with exemplar choices already made by the scorer:

```
(Frame s1_f1 lend)
(Role s1_f1 agent alex)
(Role s1_f1 goal sam)
(Role s1_f1 theme cam1)
(isa cam1 Camera)
(has-prop cam1 Nikon)
(nearest-ex cam1 Camera professional_camera)

(Frame s2_f1 return)
(Role s2_f1 agent sam)
(Role s2_f1 theme it_s2)

(Frame s3_f1 find)
(Role s3_f1 agent alex)
(Role s3_f1 theme camX)
(isa camX Camera)
(nearest-ex camX Camera professional_camera)
(part lens1 camX)
(cracked lens1)

(Frame s4_f1 borrow)
(Role s4_f1 agent sam)
(Role s4_f1 theme lens_claim_s4)
(nearest-ex lens_claim_s4 Lens standalone_lens)
```

Note three details that matter for the weaves below. First, both `cam1` and `camX` have `professional_camera` as their nearest exemplar: the Nikon cue propagates through to "their camera" because no contrary evidence appears. Second, `lens1` in S3 is a part of the professional camera, so its implicit exemplar is `camera_lens`. Third, the lens that Casey claims Sam borrowed has a different exemplar, `standalone_lens`, because "borrowed the lens" suggests an independent piece of equipment. That single exemplar difference is the key signal in the conflict.

Weave 1: pronoun to lent camera (low cost).

```
(Weave W12 S1 S2)
(MapFrame W12 s1_f1 s2_f1)
(MapEntity W12 cam1 it_s2)
(MapExemplar W12 professional_camera professional_camera)
(WeaveCost W12 0.20)
(IdPlus cam1 it_s2 0.15
  (reasons anaphora theme-role return-after-lend exemplar-match))
(IdMinus cam1 it_s2 0.80
  (reasons time-shift))
```

The pronoun "it" has only weak intrinsic information, but the surrounding structure makes the alignment cheap: same theme role, return-after-lend pattern, no exemplar shift. The negative channel is high (it would take expensive evidence to refute), and the only real negative cue is the one-day time shift, which is normal for lending and returning. Total weave cost: 0.20.

Weave 2: "their camera" to lent camera (slightly higher cost).

```
(Weave W13 S1 S3)
(MapEntity W13 cam1 camX)
(MapExemplar W13 professional_camera professional_camera)
```

```
(WeaveCost W13 0.31)
(IdPlus cam1 camX 0.24
  (reasons camera-kind nikon-cue exemplar-match owner-context))
(IdMinus cam1 camX 0.62
  (reasons location-shift possessive-ambiguous))
```

Here the alignment is still good but two negative cues add up: the possessive "their" is genuinely ambiguous between Alex and Sam, and the location has shifted to Alex's home. Still, the kind, the Nikon cue, and the exemplar all match. Total cost: 0.31, modestly higher than W12.

Weave 3: lens to claimed-lens (high cost, guarded).

```
(Weave W34 S3 S4)
(MapEntity W34 lens1 lens_claim_s4)
(MapKind W34 Lens Lens)
(MapExemplar W34 camera_lens standalone_lens)
(ExemplarShift W34 Lens camera_lens standalone_lens 0.58)
(WeaveCost W34 0.67)
(IdPlus lens1 lens_claim_s4 0.42
  (reasons same-kind same-topic same-person-sam))
(IdMinus lens1 lens_claim_s4 0.21
  (reasons camera-lens-vs-standalone-lens conflicting-claim))
(GuardedAlternative lens1 lens_claim_s4 source-casey)
```

The exemplar shift from `camera_lens` to `standalone_lens` costs 0.58 on its own and dominates the weave. The combined cost is 0.67 — more than twice W12. The system therefore does not assert that Casey's claimed lens is the same lens as the cracked one. It records a `GuardedAlternative` tagged with the source (`source-casey`), so the query planner can keep both interpretations open.

The resulting PLN bridges should let the system infer that Sam probably returned the lent camera, that the found camera is probably the same professional camera, and that the borrowed-lens claim is relevant but conflicting. The last claim should lower confidence or create an alternative; it should not make the whole inference chain explode.

9.2 Example 2: rule transfer from chess-game context to football-game context

Suppose the source context says:

> The game required deep strategy. Strategic games require patience.

and the target context says:

> The game was physically exhausting.

The word "game" is canonicalized to the same kind, but the active exemplars differ:

```
(Frame sA_f1 require)
(Role sA_f1 theme gameA)
(Role sA_f1 object strategy)
(isa gameA Game)
(nearest-ex gameA Game chess_game)
(exemplar gameA Game chess_game 0.05)
(exemplar gameA Game football_game 0.80)

(Frame sB_f1 exhaust)
```

```
(Role sB_f1 theme gameB)
(isa gameB Game)
(nearest-ex gameB Game football_game)
(exemplar gameB Game football_game 0.06)
(exemplar gameB Game chess_game 0.82)
```

The kind matches and a naive parser would treat these as the same symbol. But the exemplar distances tell a different story: `gameA` is firmly in chess territory ($d = 0.05$) and `gameB` is firmly in football territory ($d = 0.06$). Bridging them requires either a direct exemplar shift or a detour through a generic exemplar.

The direct weave. A direct chess→football mapping is possible, but expensive:

```
(Weave W_game_direct SA SB)
(MapKind W_game_direct Game Game)
(MapEntity W_game_direct gameA gameB)
(MapExemplar W_game_direct chess_game football_game)
(ExemplarShift W_game_direct Game chess_game football_game 0.80)
(WeaveCost W_game_direct 0.91)
```

The direct exemplar shift contributes 0.80; the remaining 0.11 comes from minor frame and role mismatches between `require`-strategy and `exhaust`.

The generic-game weave. A detour through a generic-game exemplar may be better when one wants a broad but weak transfer:

```
(Weave W_game_generic SA SB)
(MapKind W_game_generic Game Game)
(MapEntity W_game_generic gameA gameB)
(MapExemplar W_game_generic chess_game generic_game)
(MapExemplar W_game_generic generic_game football_game)
(ExemplarShift W_game_generic Game chess_game generic_game 0.35)
(ExemplarShift W_game_generic Game generic_game football_game 0.35)
(WeaveCost W_game_generic 0.74)
```

Two shorter shifts ($0.35 + 0.35 = 0.70$, plus 0.04 for the frame mismatch) give total cost 0.74 — somewhat cheaper than the direct route. This expresses "the analogy goes through a more abstract notion of game" explicitly, rather than pretending the two games are interchangeable.

Tracing the transported rule. A source rule with truth value $(0.85, 0.70)$:

```
(Rule strategic-games-require-patience
  (Ante (and (isa $g Game)
             (nearest-ex $g Game chess_game)
             (requires $g strategy)))
  (Cons (requires $g patience))
  (stv 0.85 0.70))
```

transported through `W_game_direct` (cost 0.91) with $\lambda_s = \lambda_w = 1.0$ produces:

$$s' = 0.85 \cdot e^{-0.91} \approx 0.34,$$
$$w' = \frac{0.70}{1 + 0.91} \approx 0.37.$$

The transported rule:

```
(TransportedRule W_game_direct strategic-games-require-patience
  (Rule weak-football-analogy
    (Ante (and (isa $g Game)
               (nearest-ex $g Game football_game)))
    (Cons (possibly-requires $g patience))
    (stv 0.34 0.37)))
```

The numbers above use the formula given in Âğ7.4. The original draft used somewhat different parameters; what matters is that the calibration is principled and traceable to the weave cost.

The key distinction is that the parser has not merely matched the word "game". It has represented a costly exemplar shift from chess-like game to football-like game. This makes the analogy usable but weak, with the strength quantitatively tied to the exemplar distance.

9.3 Example 3: normalized meaning graph alignment

For longer RAG chunks, the same mechanism aligns graph parts rather than just sentence frames. Let G_A and G_B be SENF-derived meaning graphs. A TransWeave is then a sparse coupling:

$$\Pi_{ij} \geq 0, \quad \sum_j \Pi_{ij} \leq 1, \quad \sum_i \Pi_{ij} \leq 1.$$

For each pair with $\Pi_{ij} > \theta$, copy or rewrite part of $G_A[i]$ into $G_B[j]$ with mass Π_{ij} and with confidence degradation determined by edit cost and exemplar shift:

$$\mathsf{stv}(s, w) \mapsto \mathsf{stv}\left(s \exp(-\lambda C_{ij}), \frac{w}{1 + \mu C_{ij}}\right).$$

This gives a practical version of "transporting normalized meaning graphs". It can be implemented with graph edit distance, role-graph matching, or a learned pair scorer, then exposed to PLN as explicit bridge links. The sparsity constraint $\sum_j \Pi_{ij} \leq 1$ is what prevents the same source node from being duplicated wholesale into many target nodes — in transport-language, mass is conserved.

10 How query planning should use the new layer

The current query planner already ranks executable candidates using signals such as predicate alignment, argument grounding, variable-binding feasibility, and available fact signatures. The extension should add identity and exemplar features:

$$\begin{aligned} Score(q) = {} & u_1 Score_{exec}(q) + u_2 Score_{ground}(q) + u_3 Score_{fact}(q) \\ & + u_4 Support_{Id^+}(q) + u_5 Coherence_{Ex}(q) - u_6 Penalty_{Id^-}(q) - u_7 Penalty_{transport}(q). \end{aligned}$$

The planner should prefer queries that can bind through low-cost identity paths and coherent exemplar alignments.

For the camera example, the planner should prefer

```
(EvaluationLink returned sam cam1)
```

over a literal query involving `it_s2`, because the identity graph has a strong low-cost path from `it_s2` to `cam1`. The new features make this preference principled rather than ad hoc: `cam1` is preferred because $Support_{Id^+}$ is high (the IdPlus edge has cost 0.15), $Coherence_{Ex}$ is high (both exemplars are `professional_camera`), and $Penalty_{Id^-}$ is low. A query that bound to a different camera mention from a different chunk would lose on every one of those signals.

11 Use of LangExtract-style extraction

The benchmark results showed that LangExtract-based extraction produced over-literal symbolic representations that hurt symbol reuse, with 25 out of 25 cases falling back to canonical parsing. The lesson is not that extraction is useless — it is that extraction should not own symbol generation. In the proposed layer, extraction supplies features for SENF fields rather than creating symbols of its own:

- candidate entities;
- role hints;
- time and location hints;
- source spans;
- property hints;
- possible exemplar cues;
- modality and uncertainty cues.

Then `canonical_pln` continues to own symbol normalization. This lets structured extraction contribute semantic signal without damaging symbolic reuse. In other words, LangExtract becomes a feature provider for the SENF builder rather than a parallel parsing path.

12 MVP implementation plan

A minimal extension can be built in six phases. Each phase produces a testable artifact; later phases depend on earlier ones, so the team can stop at any point and still have a usable system.

12.1 Phase 1: SENF builder

Convert canonical PLN atoms into SENF objects. This phase should be deterministic where possible and should preserve source spans and canonical symbol names.

12.2 Phase 2: exemplar scorer

For each entity-kind pair, score distances to a small exemplar set. Keep the top few exemplar hypotheses under guards. The scorer can start as a simple lookup table over lexical cues plus an LLM prompt restricted to canonical symbols; better embeddings can be added later.

Algorithm 3 Build SENF from canonical PLN atoms

Require: Canonical PLN atom set A and source spans P

Ensure: SENF S

Create empty frame, entity, role, kind, constraint, and graph sets.
for all atom $a \in A$ **do**
    Classify a as frame, role, kind assertion, property assertion, rule, or constraint.
    Add canonical symbols and source-span evidence to the appropriate SENF field.
    Preserve parser confidence and source modality as guards.
end for
Add default entity mentions for unbound but role-bearing arguments.
Add initial identity edges for exact canonical matches and protected constants.
Return the populated SENF.

12.3 Phase 3: identity graph

Build identity edges across mentions in a local sentence window, then across retrieved chunks. Store reasons and negative evidence. The first version can use rule-based scoring; a learned scorer can be added later once the harness produces enough labeled identity pairs.

12.4 Phase 4: TransWeave

Build top-k weaves between relevant SENF pairs: sentence-to-sentence, chunk-to-query, chunk-to-chunk, and source-rule-to-target-context.

The recommended structure is the hierarchical TransWeave of Âğ8.4, built in two stages. *Stage 4a* produces the heuristic pushforward seed using canonical kind matches, lexical aliases, anaphora cues, and role positions — almost all of the signal needed is already computed by Phases 1–3, so this stage is mostly assembly work. *Stage 4b* adds the local Sinkhorn-style polish over candidate pairs, then block polish over short frame windows, then a small global polish. For very small SENF pairs (single sentence to single sentence) the block stage can be skipped and the system degrades cleanly to local-only polish.

A simpler single-shot algorithm (Hungarian matching on entity and frame pairs) is fine as the first cut for the smallest cases and as a baseline for evaluating the hierarchical version; beam-search and ILP variants of the single-shot algorithm can come later if needed.

12.5 Phase 5: PLN bridge generation

Emit bridge links such as:

```
(SimilarityLink cam1 it_s2 (stv 0.84 0.72))
(ContextLink (Exemplar Camera professional_camera)
  (SimilarityLink cam1 it_s2 (stv 0.84 0.72)))
(ImplicationLink
  (EvaluationLink damaged lens1)
  (EvaluationLink damaged camX)
  (stv 0.66 0.50))
```

These atoms are what the PLN reasoner actually consumes, so this phase is where the new layer becomes visible to the rest of the pipeline.

12.6 Phase 6: query planner integration

Add identity/exemplar coherence terms to query ranking and include weave explanations in parser diagnostics. The diagnostics are important: they let the team understand why a proof was accepted or guarded, and they make the benchmark failure modes interpretable.

13 Evaluation plan

Use the existing benchmark categories and add identity-specific tests.

Existing operational metrics:

- proof discovery;
- query alignment;
- fallback usage;
- no-query events;
- latency.

New identity-aware metrics:

- pronoun resolution accuracy;
- definite-description resolution accuracy;
- exemplar-choice accuracy;
- contradiction containment rate;
- transported-rule calibration (does the truth value of a transported rule track its actual reliability?);
- percentage of proofs that rely on bridge links;
- number of false positive bridge links.

The desired outcome is not just more proofs. It is more proofs with fewer brittle symbol mismatches, fewer no-query failures, and better explanations of why a proof was accepted or guarded. A useful sanity check: the proof-rate improvement should come disproportionately from the cases that the current `canonical_pln` fails on, not from easy cases it already solves.

14 What not to implement first

Do not begin with:

- a full HoTT proof assistant;
- full univalence machinery;
- infinity-groupoid structures;

- a large manually built ontology;
- global consistency repair;
- unrestricted LangExtract phrase symbols.

Begin with:

- small exemplar sets;
- local identity graphs;
- top-k weighted TransWeaves;
- PLN bridge links;
- confidence degradation tied to weave cost;
- query planner features that use the new signals;
- diagnostics showing the reasons for identity and non-identity.

The discipline here mirrors the lesson from the canonicalization workstream: tight, deterministic, debuggable mechanisms beat ambitious but unstable ones.

A Appendix: quick HoTT background for interested implementers

The main body does not require HoTT knowledge. This appendix explains how the practical objects correspond to the mathematical ideas, for readers who want to know what the implementation is approximating.

A.1 Map between HoTT objects and engineering objects

HoTT object	Engineering object	Where it appears in the implement
Identity type $\mathsf{Id}_A(x, y)$	Identity edge	Identity graph; `IdPlus` edges
Graded identity $\mathsf{Id}^{+}_{\leq\tau}(x, y)$	Identity edge with cost	`IdPlus` edge with c^{+}
Negative graded identity $\mathsf{Id}^{-}_{\leq\sigma}$	Identity edge, negative channel	`IdMinus` edge with c^{-}
Quantale (cost algebra)	Real-valued costs with addition	Cost field on edges and weaves
Multi-exemplar contractibility	Exemplar set per kind	Exemplar registry
Definite description "the N"	Nearest-exemplar selection	`sel` and `nearest-ex`
Univalence / equivalence as path	Bounded weave between contexts	`Weave` object with cost $C(W)$
Bellman-Darboux intertwining	Approx. commuting with discourse step	Marginal constraints in Sinkhorn p
Discourse Bellman operator B_c	Discourse / inference step operator	Implicit; never built explicitly
Intertwining error ε	Polish residual at each scale	$(\varepsilon_{loc}, \varepsilon_{blk}, \varepsilon_{glob})$ diagnostics
Transport along a path	Symbol rewrite + truth degrade	PLN bridge generator
Modalities ($\flat$, $\sharp$, Cons)	Forget costs / threshold / world tag	Optional planner modes
Paraconsistency	Both Id^{+} and Id^{-} kept	Negative channel does not erase po

The rest of this appendix expands each row.

A.2 Identity as paths

In ordinary first-order symbolic systems, identity is a relation $x = y$, true or false. In HoTT, identity is a type of paths:

$$\mathsf{Id}_A(x, y).$$

An inhabitant of this type is concrete evidence that x and y are the same in type A. The practical identity graph implements this idea by storing explicit evidence paths between mentions: the edge *is* the inhabitant, and the reasons attached to it are the proof.

A.3 Graded identity

PGM-HoTT adds costs or grades to identity evidence:

$$\mathsf{Id}^{+}_{A,\leq\tau}(x, y),$$

meaning there is positive evidence for identifying x and y within budget τ. The implementation represents this as an `IdPlus` edge with a cost c^+ and a set of reasons.

PGM-HoTT also has a negative channel:

$$\mathsf{Id}^{-}_{A,\leq\sigma}(x, y),$$

meaning there is positive evidence for difference or non-identity. The implementation represents this as an `IdMinus` edge with a cost c^-. This is paraconsistent: the system has both sameness evidence and difference evidence on the same pair without collapsing, because classical negation is never invoked.

A.4 Quantales as cost algebras

A quantale is an algebra for composing costs. The implementation uses the simplest one: costs are nonnegative real numbers and composition is addition, satisfying the triangle inequality

$$c(x, z) \leq c(x, y) + c(y, z).$$

An indirect identity path through y gives an upper bound on the cost of identifying x with z. In implementation terms, identity propagation can use shortest-path or top-k path search.

A.5 Multi-exemplar contractibility

Classical HoTT treats definite descriptions through contractibility: a type is contractible if it has one canonical center and every element is identical to that center. PGM-HoTT generalizes this to multiple centers or exemplars.

For a kind K, instead of one prototype, use a finite exemplar set:

$$X_K = \{x_1, \ldots, x_n\}.$$

Each entity e has distances to the exemplars:

$$d(e, x_1), \ldots, d(e, x_n).$$

The practical exemplar registry is a direct implementation of this multi-exemplar structure.

A.6 The meaning of "the"

In classical HoTT, "the N" is licensed only when the type of N-witnesses is contractible (one canonical center). In PGM-HoTT, "the N" is contextual nearest-exemplar selection. The implementation therefore resolves "the camera", "the game", or "the restaurant" by using context, role structure, and exemplar distance:

$$\mathsf{the}_{multi}(K, y) = \operatorname*{argmin}_{x \in X_K} d(y, x),$$

with guarded alternatives when several exemplars are close.

A.7 Transport

In HoTT, if we have a dependent family $P : A \to \mathsf{Type}$ and a path $p : x = y$, then transport moves an inhabitant of $P(x)$ to an inhabitant of $P(y)$:

$$\mathsf{transport}_P(p) : P(x) \to P(y).$$

In the implementation, a TransWeave is the path-like object. Transporting a fact or rule through a weave means rewriting its symbols through the weave's maps $(\phi_E, \phi_K, \phi_X, \phi_R)$ and degrading its PLN truth value according to the weave cost $C(W)$.

A.8 Univalence

Univalence says, roughly, that equivalent types can be identified. The operational reading is useful: if two contexts have a sufficiently good equivalence, then semantic objects can be moved between them. In the implementation,

$$W : S_A \rightsquigarrow S_B$$

acts as a bounded equivalence between two semantic contexts. A low-cost weave licenses transport with little confidence loss. A high-cost weave produces a weak analogy or a guarded alternative.

A.9 Modalities

Modalities change perspective. For the implementation, the useful modalities are:

- forget costs ($\flat$): use the coarse canonical identity only;
- enforce a consistency threshold ($\mathsf{Cons}_{\leq \tau}$): ignore or quarantine negative evidence below a chosen bound;

- search under a cost budget ($\Diamond_{\leq r}$): find alignments with cost at most r;
- store world or context labels: reason inside a sentence, chunk, source, or task context.

The MeTTa-IL direction (discussed in the Modal-HoTT-MeTTa note) can later expose these as cost-indexed reachability and bilattice operators. This should come after the SENF/TransWeave prototype works, not before.

B Appendix: formalizing the main body in HoTT terms

This appendix gives the formal reading of each engineering object. It is for readers who want to see how the implementation lines up with the mathematical framework; the implementation itself does not depend on it.

B.1 Contexts and fibers

Let Ctx be a type or category of contexts. A context may be a sentence, chunk, source, task, or world. Define a dependent family:

$$\mathsf{Sem} : \mathsf{Ctx} \to \mathsf{Type}.$$

An element of $\mathsf{Sem}(c)$ is a semantic object valid in context c, such as a SENF, a PLN assertion set, or a proof state.

B.2 SENF as an inhabitant of a semantic fiber

For a sentence or chunk context c, its normalized representation is:

$$S_c : \mathsf{Sem}(c).$$

The fields of S_c are practical projections:

$$F(S_c), E(S_c), K(S_c), X(S_c), R(S_c), C(S_c), G(S_c).$$

The identity graph $G(S_c)$ stores approximate inhabitants of graded identity types.

B.3 Identity edges as graded paths

A practical edge

```
(IdPlus e1 e2 0.18 (reasons ...))
```

corresponds to a low-cost inhabitant of:

$$\mathsf{Id}^{+}_{A,\leq 0.18}(e_1, e_2).$$

A practical edge

```
(IdMinus e1 e2 0.21 (reasons ...))
```

corresponds to evidence in the negative identity channel. The system can have both without explosion because the channels are not forced into classical negation.

B.4 Exemplar registry as finite centers

For a kind K, the exemplar registry gives:

$$X_K : \mathsf{FiniteSet}(K).$$

The distance assertions

```
(exemplar e K x d)
```

are practical forms of geodesic distances from e to exemplar x.

B.5 TransWeave as a graded equivalence or morphism

A weave between contexts c_A and c_B provides a partial, graded equivalence:

$$e_W : \mathsf{Sem}(c_A) \simeq_{C(W)} \mathsf{Sem}(c_B).$$

By a guarded form of univalence, such an equivalence can be treated as a path in a universe of semantic fibers:

$$p_W : \mathsf{Sem}(c_A) = \mathsf{Sem}(c_B).$$

Then HoTT transport gives:

$$\mathsf{transport}_{\mathsf{Sem}}(p_W) : \mathsf{Sem}(c_A) \to \mathsf{Sem}(c_B).$$

The implementation approximates this by mapping frames, roles, entities, kinds, and exemplars through W, then degrading PLN truth values by $C(W)$.

This subsection describes what a weave *produces* (an equivalence, a path, a transport map). The next subsection describes what makes a weave *a weave* rather than an arbitrary structured map: the Bellman commutation that ties the equivalence to the dynamical operators on each side.

B.6 TransWeave as Bellman-Darboux intertwining

This is the mathematical content that the rest of the appendix has so far skipped: the commutation condition that defines TransWeave and makes the word "weave" literal rather than metaphorical.

The discourse Bellman operator. For each context c (sentence, chunk, source, task), let $B_c : \mathsf{Sem}(c) \to \mathsf{Sem}(c)$ be the operator that propagates SENF state by one step. The exact form depends on what counts as a step:

- *Per-sentence discourse*: B_c takes the SENF state after sentence n and produces the SENF state after sentence $n+1$ — it adds new mentions, extends the identity graph, updates exemplar nearest-neighbors, and resolves anaphora that the new sentence makes possible.
- *PLN inference*: B_c takes a set of atoms with truth values and applies one step of inference — deduction, induction, abduction, revision.

- *Chunk-level retrieval*: B_c takes the current evidence set and incorporates the next retrieved chunk's contribution.

In each case B_c is the dynamical operator that defines how the context evolves. The TransWeave does not need to know the internal details of B; what it needs is that B exists and that two such operators (B_A for context A, B_B for context B) can be compared.

The intertwining condition. A mapping $W : \mathsf{Sem}(c_A) \to \mathsf{Sem}(c_B)$ is a TransWeave with intertwining error ε when

$$\|W \circ B_A - B_B \circ W\| \leq \varepsilon$$

in an appropriate operator norm (typically a sup-norm over reachable SENF states under a salience-weighted measure). The diagram

$$\begin{array}{ccc} \mathsf{Sem}(c_A) & \xrightarrow{B_A} & \mathsf{Sem}(c_A) \\ \downarrow W & & \downarrow W \\ \mathsf{Sem}(c_B) & \xrightarrow{B_B} & \mathsf{Sem}(c_B) \end{array}$$

commutes up to ε. In plain English: doing a step of discourse evolution in A and then weaving over to B produces the same SENF state (up to ε) as weaving over first and then doing a step of discourse evolution in B.

This is what the word "weave" is meant to convey. The map W weaves together two dynamical systems — two contexts, each with its own way of propagating meaning — in a way that respects their dynamics. A correspondence that did not commute with the operators would not be a weave; it would just be a renaming.

Why this matters for PLN transport. The commutation has a direct operational consequence:

- If ε is small, transporting a fact through W and then doing PLN inference gives approximately the same answer as doing PLN inference first and then transporting. Order does not matter much; the system can defer or anticipate context shifts freely.
- If ε is large, order matters. Doing inference first and then transporting can disagree substantially with transporting first and then inferring. The system must commit to one order, and the disagreement is the calibration cost the PLN bridge generator must absorb.

The total weave cost $C(W)$ is what bounds ε. The exponential confidence-degradation formula $s' = s\exp(-\lambda C(W))$ from Âğ8.5 is essentially passing ε through to truth-value discounting.

Connection to the transport trinity. The PGM-HoTT note decomposes a Bellman-Darboux transport cost as

$$d_{TW} = \alpha\, d_{struct} + \beta\, d_{dist} + \gamma\, d_{dyn},$$

corresponding to structural (Gromov-Wasserstein-like), distributional (Wasserstein-like over value distributions), and dynamic (SchrÃűdinger-bridge-like over evolution) transports. The Âğ7.4 structural/exemplar/conflict decomposition is the parser-side instance of the same trinity: *structural* corresponds to role-pattern and frame-shape preservation, *distributional* corresponds to exemplar shifts under d_X, and *dynamic* corresponds to conflict between the IdMinus channels and the action of B. The intertwining error ε is bounded by a weighted sum of these three components.

Why hierarchical Sinkhorn enforces commutation at the right scales. The hierarchical polish in Âğ8.4 is exactly the procedure that drives ε down at each scale. The connection is:

- *Local polish.* For each candidate pair (e, e'), the row and column marginals that Sinkhorn/IPF enforces encode the one-step commutation. Concretely: if Π_E couples a source mention e to a target mention e', the row marginal $\sum_{e'} \Pi_E(e, e') = \mu_A(e)$ says "the total mass leaving e under W equals e's salience in S_A." The column marginal says the same on the target side. Together they encode the equation

$$W(B_A(\delta_e)) \approx B_B(W(\delta_e))$$

  on a single-step neighborhood. The k_0 Sinkhorn sweeps drive the local residual below tolerance.

- *Block polish.* For a block of W consecutive discourse steps, the block boundary marginals encode the W-step commutation

$$W(B_A^W(s)) \approx B_B^W(W(s))$$

  where B^W denotes W-fold composition. The Sinkhorn sweeps at block boundaries drive this residual down; redistributing corrections back to the local couplings keeps single-step commutation intact.

- *Global polish.* The final k_2 sweeps enforce commutation over the full discourse horizon, including the terminal goal T_G: the SENF state at the end of c_A should map under W to the SENF state at the end of c_B, up to the global intertwining bound.

This is why the hierarchical algorithm is not just a computational convenience. It is the natural way to enforce the Bellman commutation at multiple scales simultaneously: small residuals locally, accumulated and corrected at the block scale, with a final global check.

Connection to Bellman-Darboux in the PGM-HoTT note. The PGM-HoTT note defines a BD (Bellman-Darboux) transformation $T : M_1 \to M_2$ between MDP-like contexts as a map satisfying $\|T_V \circ B_{M_2} - B_{M_1} \circ T_V\|_\infty \le \varepsilon$, where T_V is the value pullback and B_{M_i} is the value-iteration Bellman operator. The TransWeave construction in the parser setting is exactly this with M_i being discourse contexts rather than MDPs, B_{M_i} being a discourse-update operator rather than a value-iteration operator, and the transport trinity becoming the structural/exemplar/conflict decomposition. The HoTT transport that lifts the fibered equivalence to a path in the universe of semantic fibers is the same construction in both settings; the implementation difference is only that the parser's Bellman operator is discrete-discourse rather than continuous-MDP.

Practical implication for the implementation team. The team does not need to construct B_A and B_B explicitly or measure ε directly. The Sinkhorn marginal constraints inside the local, block, and global polish stages are sufficient: enforcing the marginals *is* approximately enforcing the commutation. The intertwining condition is a target the polish converges toward, not a side computation the system has to perform.

What the team should expose is a way for the diagnostics layer to report the per-scale residuals — how far the local polish leaves marginals unsatisfied, how far the block polish, how far the global polish. These residuals are direct estimates of ε at each scale, and they are the right signal for telling a user (or another component) “this weave is well-formed” vs. “this weave is being forced and you should not trust transported truth values from it.” A simple summary tuple $(\varepsilon_{loc}, \varepsilon_{blk}, \varepsilon_{glob})$ printed alongside $C(W)$ and $\rho(W)$ gives the rest of the system everything it needs to interpret the weave’s reliability.

B.7 PLN bridge generation as cost-consuming transport

A PLN assertion a with truth value (s, w) in $\mathsf{Sem}(c_A)$ is transported to a target assertion a' in $\mathsf{Sem}(c_B)$:

$$a' = W(a),$$

with truth value:

$$(s', w') = \left(s \exp(-\lambda C(W)), \frac{w}{1 + \mu C(W)}\right).$$

This is the implementation-level version of graded transport: using the path consumes cost.

B.8 Guarded alternatives as unresolved inhabitants

When two interpretations have close costs, the system should not collapse them. It should store:

```
(GuardedAlternative alt1 guard1 cost1)
(GuardedAlternative alt2 guard2 cost2)
```

Formally, these are inhabitants available under guards or modalities. Operationally, the query planner can choose one under a budget, or PLN can carry both with different truth values.

B.9 Query planning as cost-bounded proof search

The identity-aware query planner can be read as cost-bounded search for a path from a query representation to executable known facts and rules. In HoTT terms, it searches for a low-cost transport path plus a proof in the target fiber. In implementation terms, it ranks candidate queries by executable alignment, identity support, exemplar coherence, and transport cost.

C Appendix: minimal API contract

A small implementation API could expose the following operations:

```
(build-senf canonical-pln-atoms source-spans) -> SENF
(score-exemplars SENF exemplar-registry) -> SENF
(build-identity-graph SENF context-window) -> IdentityGraph
(build-weaves source-SENF target-SENF k) -> (Weave ...)
(emit-pln-bridges Weave) -> (PLNAtom ...)
(rank-queries-with-weaves query-candidates identity-graph weaves) -> RankedQueries
```

This API is intentionally small. It gives the team a practical path toward the HoTT ideas while allowing the mathematical machinery to remain in the background, where it belongs for an engineering deliverable.

References

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